

A Theory of International Boycotts

Abdoulaye Ndiaye

NYU Stern School of Business

SED Meetings 2025, Copenhagen



Motivation

- ▶ **Geoeconomics:** International boycotts are significant in international trade and geopolitical relations.
- ▶ **Sanctions?** Unlike government-imposed sanctions and tariffs, collective choice of consumers.
- ▶ Broadly *effects of strategic shifts in consumer preferences* for international trade
- ▶ Key Questions
 - ▶ How do international boycotts impact firms and nations?
 - ▶ What distinguishes the effects of boycotts from those of sanctions or tariffs?
 - ▶ What are the efficiency and equity considerations in boycotts?

Objective

- ▶ This paper studies the economics of international boycotts within a **neoclassical** framework.
- ▶ Studies optimal targeted boycott.
- ▶ Key Findings
 - ▶ Boycotts can operate on the consumption side or labor supply side
 - ▶ **Consumption side:** Boycotts can be more effective as ban on goods near comparative advantage barrier.
 - ▶ **Labor supply side:** Boycotters optimally work more to spend on domestic goods
 - ▶ Some **symmetry** btw boycotts and VERs but **leakage** from nonboycotters.

Facts on the Impact of Consumer Boycotts

Long history of boycotts.

- ▶ With globalization, "**surrogate**" boycotts intensify, citizens target foreign firms due to mainly geopolitical tensions
 - ▶ **US Boycott of French wine (Early 2000s)** due to France's opposition to the Iraq war, resulting in a \$112 million sales decrease.
 - ▶ **Boycotts of Danish (2015) and French (2015) products** by Arab league.
- ▶ Government branches sometimes support consumer boycotts (Bohman 2022, Hong et al. 2011)
 - ▶ **Chinese boycott of French cars (2008)** for the support expressed for Tibet, resulting in a loss of market share between 25% and 33%
 - ▶ **Boycotts following statements opposing the Chinese government**, which in 50% of cases cause targets to publicly apologize.

Model

Other Aspects of Boycotts

Conclusion

Model Overview

- ▶ Theoretical framework based on the Dornbusch, Fischer, and Samuelson (1977).
- ▶ Two countries, Home (H) and Foreign (F).
- ▶ Trade occurs over a continuum of goods indexed by $j \in [0, 1]$.
- ▶ Production
 - ▶ Single input: labor (l), with constant marginal product for each good j .
 - ▶ Country-specific productivity: $z_H(j)$ for Home and $z_F(j)$ for Foreign.
 - ▶ Relative productivity of Home to Foreign: $Z(j) = \frac{z_H(j)}{z_F(j)}$.
 - ▶ Labor endowments: l_H and l_F , with free labor movement within countries.

Regular Consumers

Consider regular consumers in both countries:

$$U(\mathbf{q}) = \int_0^1 s(j) \log(q^i(j)) dj \quad (1)$$

Constraints:

$$\int_0^1 p(j) q^i(j) dj = w_i l_i \quad (2)$$

- ▶ $q^i(j)$: quantity of good j consumed.
- ▶ $s(j)$: expenditure share on good j , $S(1) = 1$.
- ▶ Non-substitution theorem:
 - ▶ prices uniquely determined by supply side $p(j; \{w_H, w_F\})$
 - ▶ utilities are determined by vector of world wages $U^H(\{w_H, w_F\})$ and $U^F(\{w_H, w_F\})$

Boycotting Coalition Problem

Exogenous hare λ of domestic boycotters, behave so as to maximize their own welfare subject to maintaining given welfare level v^F of foreign households.

$$U^B(\{w_H, w_F, v^F\}) = \max_{(q_H^B, q_F^B) \in X} \int_0^1 s(j) \log(q_H^B(j) + q_F^B(j)) dj \quad (3)$$

$$\text{s.t: } \int_0^1 p_H(j) q_H^B(j) dj + \int_0^1 p_F(j) q_F^B(j) dj = w_H l_H \quad (4)$$

$$\text{broadly s.th: } U^F(\{w_H, w_F\}) \leq v^F \quad (5)$$

Competitive Boycotting Equilibrium

Definition (Competitive Boycotting Equilibrium)

A CE with boycotters corresponds to a pair of wages, $\{w_H, w_F\}$, a schedule of world prices $\mathbf{p}$, and domestic prices $\{\mathbf{p}_H, \mathbf{p}_F\}$, consumption schedules $\{\mathbf{q}^H, \mathbf{q}^F, \mathbf{q}^B\}$, foreign utility v^F , and output schedules that satisfy:

1. Firm optimization: price equal firms' marginal cost,
2. Regular consumer utility maximization: domestic and regular consumers maximize (1) subject to (2),
3. Foreign utility constraint: $U^F(\{w_H, w_F\}) \leq v^F$
4. Labor market clearing in each country

$$\int_0^1 \frac{((1 - \lambda)q^H(j) + \lambda q_H^B(j) + q^F(j))}{z_H(j)} dj = l_H \quad (6)$$

$$\int_0^1 \frac{((1 - \lambda)q^H(j) + \lambda q_F^B(j) + q^F(j))}{z_F(j)} dj = l_F \quad (7)$$

5. No other profile $\{w'_H, w'_F, \mathbf{p}'_H, \mathbf{p}'_F, \mathbf{q}'^H, \mathbf{q}'^F, \mathbf{q}'^B, v^F\}$ satisfies 1. 2. 3. 4. and

$$U^i(\{w'_H, w'_F\}) = U^i(\{w_H, w_F\}), i \in \{H, F\}, \text{ and } U^B(\{w'_H, w'_F\}) > U^B(\{w_H, w_F\}).$$

Two Dimensions of Boycotts

- ▶ Denote $E(\mathbf{q}_i^B) \equiv \lambda \int_0^1 p_i(j) q_i^B(j) dj$, for $i \in \{H, F\}$ total expenditure of boycotters on H and F goods
- ▶ General equivalence result that boycotting CE equivalent to boycotters maximizing their utility subject to spending budgets $\{E^H(v^F), E^F(v^F)\}$
 - ▶ under assumptions for the non-substitution theorem
- ▶ Leads to two sides of how these budget can be allocated:
 - ▶ How can boycotters allocate their **consumption** to achieve their goal?
 - ▶ How do boycotters optimally choose **labor supply**?

The Consumption Side of Boycotts

Let $S(\mathbf{q}_i^B) \equiv \int_0^1 p_i(j) q_i^B(j) dj / (w_H l_H)$ be total expenditure as share of income of a boycotter

Lemma (Set of Boycotted Goods)

In any CE with boycotters where $\bar{L}_H < l_H$ or $e(\bar{L}_H) > 0$, the set of completely boycotted Foreign goods $\mathcal{J}_B = \{j \in [0, 1]; q_F^B(j) = 0\}$ is nonempty.

Proposition (Optimal Set of Boycotted Goods)

Let $\mathcal{J}_B$ be the optimal targeted set of boycotted goods given a level of tolerated Foreign welfare v^F :

- 1. There exists no set of positive measure between $\mathcal{J}_B$ and the trade cutoff $\bar{j}$, and $\mathcal{J}_B$ is a convex set.*
- 2. There exists $j_B \in [\bar{j}; 1]$ so that $\mathcal{J}_B = [0, j_B]$ (almost everywhere) boycotts goods up to j_B .*

Comparison with Uniform Import Tariffs

Figure: (A) Equilibrium with Boycott

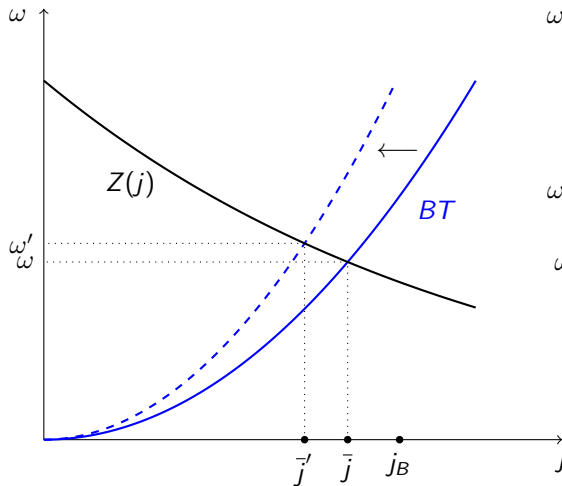
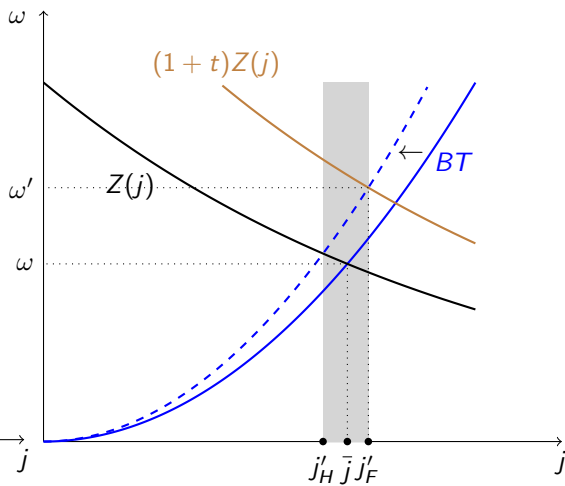


Figure: (B) Linear Import Tariffs



Leakage in the Presence of Nonboycotters $\lambda < 1$

- ▶ With the presence of nonboycotters $\lambda < 1$, imports from nonboycotters equal to

$$M_{NB} = (1 - \lambda)[1 - S(\bar{j}(j_b))]w_H l_H. \quad (8)$$

- ▶ Boycott $\uparrow w_H$.
- ▶ In addition, the export share $1 - S(\bar{j}(j_b)) \uparrow$ increases when a boycott expands.
- ▶ The combination of the two effects raises the imports of nonboycotters M_{NB}
- ▶ Related to optimal unilateral behavior by consumers w **altruistic, non-individualistic preferences**.
- ▶ Leakage pushes a boycotter to :
 - ▶ boycott a larger set of goods to tolerate the same level of Foreign welfare v^F , i.e., $j_B(\lambda, v^F)$,
 - ▶ and Foreign imports per boycotter $M_{NB} = [1 - S(j_b)]w_H l_H$ are decreasing in λ .

Symmetry in Case of Boycott by All Domestic Consumers $\lambda = 1$

- ▶ Consider case with full-boycott-coalition $\lambda = 1$
- ▶ In full-boycott-coalition economy, Home imports of foreign goods are

$$M_H = [1 - S(\mathcal{J}_b)]w_H l_H. \quad (9)$$

- ▶ $\uparrow \mathcal{J}_b \implies \downarrow 1 - S(\mathcal{J}_b)$ and indirectly $\uparrow w_H$. On net, $M_H \downarrow$

- ▶ Exports of domestic firm goods are

$$X_H = S(\bar{j}(j_B))w_F l_F. \quad (10)$$

- ▶ $\uparrow j_B \implies \downarrow \bar{j}(j_B) \implies \downarrow X_H$ Boycotts, therefore, distort both imports and exports.
- ▶ Boycott by all consumers is equivalent to a **voluntary export restrictions** by domestic firms.
- ▶ Reminiscent of **Lerner symmetry** for tariffs.

Labor Supply Side of Boycotts

Modify the DFS setup so that consumers choose consumption goods $\mathbf{q}$ and labor supply l_i to maximize

$$U(\mathbf{q}^i, l_i) = \int_0^1 s(j) u(q^i(j)) dj - v(l_i) \quad (11)$$

We consider CRRA utility for goods and an isoelastic disutility of labor:

$$u(q) = \frac{q^{1-\gamma} - 1}{1-\gamma} \quad \text{and} \quad v(l) = \frac{l^{1+\eta}}{1+\eta}, \quad (12)$$

Labor Supply Side of Boycotts

Proposition (Labor Supply of Boycotters)

In every optimal boycott, boycotters work more than regular consumers $l^{B} > l_H^*$.*

Intuition

$$V(w_i, l_i) = \frac{(w_i l_i)^{1-\gamma} P_i^\gamma - 1}{1-\gamma} - \frac{l_i^{1+\eta}}{1+\eta}. \quad (13)$$

where P_i is the price index faced by the consumer. Optimal labor supply of a boycotter

$$l^{B*} = \left[P_B^\gamma w^{1-\gamma} \right]^{\frac{1}{\eta+\gamma}}.$$

Boycotter faces a penalty in consumption basket bc must more for the goods she buys domestically.

Since $P_B > P$, $l^{B*} > l_H^*$.

Model

Other Aspects of Boycotts

Conclusion

Substitution Elasticities

- ▶ Our model extends to cases where the elasticity of substitution $\sigma \neq 1$, adopting CES preferences:

$$U(\mathbf{q}) = \left(\int_0^1 s(j)^{\frac{1}{\sigma}} q^j(j)^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}}$$

- ▶ When $\sigma \neq 1$, expenditure on each good depends on its price and the overall price index.
- ▶ Results adapt to CES preferences and more general preferences.
- ▶ Since consumers find it easier to switch to alternative products with a higher elasticity of substitution relative to that of other goods, it is **optimal to boycott goods close to $\bar{j}$ in relative productivity and substitutable with other goods**

Political Salience and Specific Firm Boycotts

- ▶ Consumers may target firms due to political salience or animus, affecting their expenditure shares.
- ▶ Preferences and choices for boycotters can include damages $d(j)$ to firms' products:

$$U^B = \max_{\{q^B\}} \int_0^1 s(j)(1 - d(j)) \log(q^B(j)) dj$$

- ▶ If the damage to imported goods is higher, the boycott improves Home's trade terms; otherwise, it harms them.

Strategic Retaliation

- ▶ Retaliation involves boycotted countries targeting the home country's exports.
- ▶ Strategic interactions can lead to a prisoner's dilemma:

$$\omega = \frac{1 - \lambda^F}{1 - \lambda^H} \cdot \frac{S(\bar{j})}{1 - S(\bar{j})} \cdot \frac{l_F}{l_H}$$

- ▶ Utilitarian governments may find no boycotts ($\lambda^H = \lambda^F = 0$) as the only Nash equilibrium.
- ▶ So boycotts need to be motivated beyond internal equity considerations.

Model

Other Aspects of Boycotts

Conclusion

Conclusion

- ▶ I developed a Ricardian model of boycotts in international trade
- ▶ Boycotts can operate on the consumption side or labor supply side
- ▶ **Consumption side:** Boycotts can be more effective as ban on goods near comparative advantage barrier.
- ▶ **Labor supply side:** Boycotters optimally work more to spend on domestic goods
- ▶ Some **symmetry** btw boycotts and VERs but **leakage** from nonboycotters.
- ▶ Some open (empirical) questions.
 - ▶ How to link measures consumption/labor supply effects to welfare?
 - ▶ How large are political salience/other considerations relative to efficiency for boycotts.
 - ▶ How large are potential leakages and threats of retaliation?

THANKS!



Comments, complaints, paper boycotts: an1490@nyu.edu